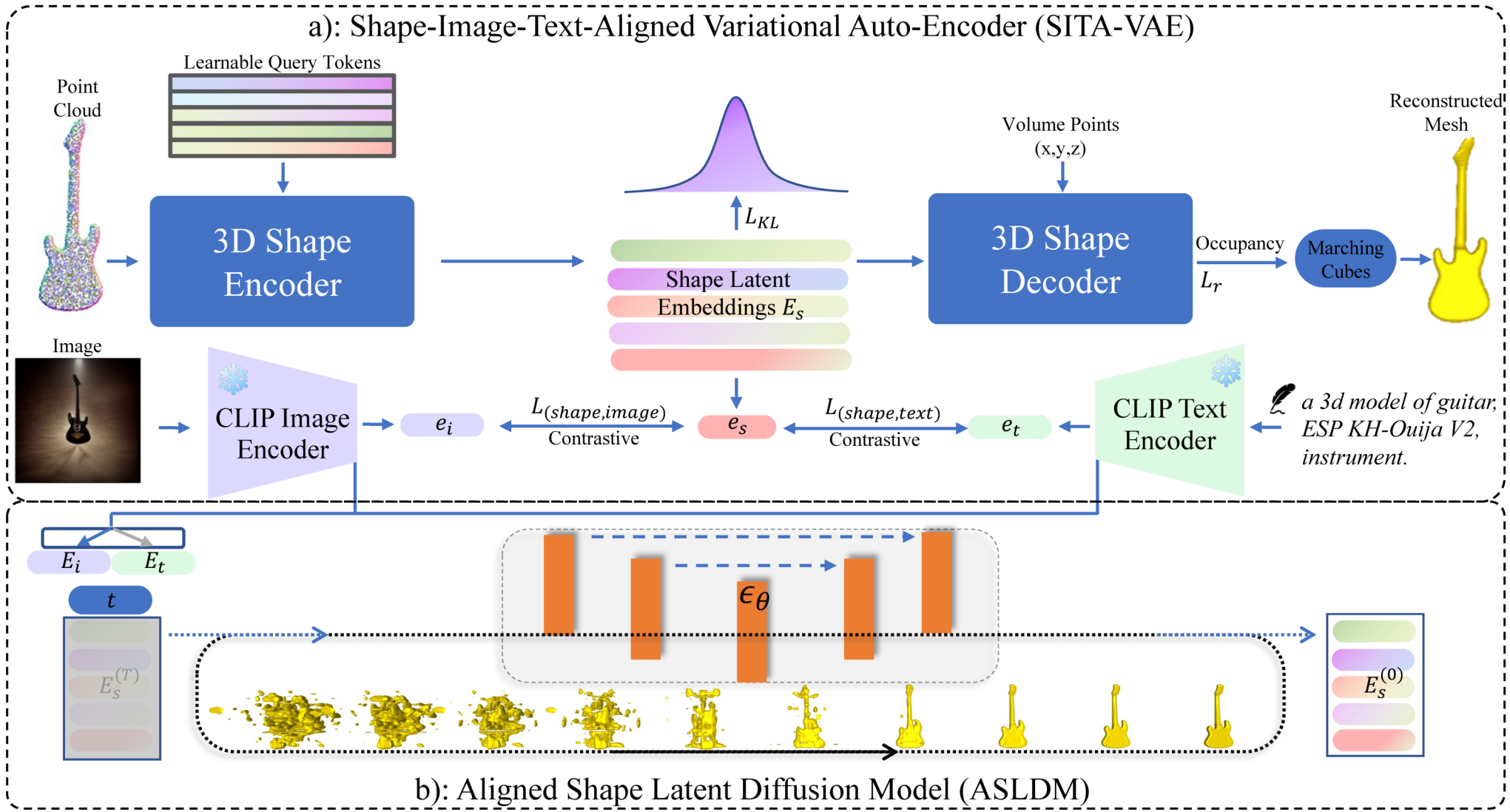
a): Shape-Image-Text-Aligned Variational Auto-Encoder (SITA-VAE)
Point Cloud
Learnable Query Tokens
3D Shape Encoder
L_{KL}
Shape Latent Embeddings E_s
Volume Points (x,y,z)
3D Shape Decoder
Occupancy
L_r
Marching Cubes
Reconstructed Mesh
Image
CLIP Image Encoder
e_i
$L_{(shape,image)}$
Contrastive
e_s
$L_{(shape,text)}$
Contrastive
e_t
CLIP Text Encoder
a 3d model of guitar, ESP KH-Ouija V2, instrument.
E_i
E_t
t
$E_s^{(T)}$
ϵ_θ
$E_s^{(0)}$
b): Aligned Shape Latent Diffusion Model (ASLDM)